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THEME SECTION PAPER

A participative end-user method for multi-perspective business process elicitation and improvement

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Abstract A business process can be characterized by multiple perspectives (intentional, organizational, operational, functional, interactional, informational, etc). Business process modeling must allow different stakeholders to analyze and represent process models according to these different perspectives. This representation is traditionally built using classical data acquisition methods together with a process representation language such as BPMN or UML. These techniques and specialized languages can easily become hard, complex and time consuming. In this paper, we propose ISEA, a participative end-user modeling approach that allows the stakeholders in a business process to collaborate together in a simple way to communicate and improve the business process elicitation in an accurate and understandable manner. Our approach covers the organizational perspective of business processes, exploits the information compiled during the elicitation of the organizational perspective and touches lightly an interactional perspective allowing users to create customized interface sketches to test the user interface navigability and the coherence within the processes. Thus, ISEA can be seen as a participative end-user modeling approach for business process elicitation and improvement.

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1 Introduction

Business Process Management is an important best practice that is critical for the long-term success of an organization and provides important benefits to organizations [27]. Modeling business processes may have different goals: align the organizational processes with users' needs, explain or automate the different processes and evolve the conduct of the business in order to adapt it more rapidly to change, etc.

Many authors agree on the fact that business process modeling techniques must enable the different stakeholders to analyze and represent business processes according to different and adapted perspectives (intentional, functional, organizational…) [12,13,21,28,29,39]. The goal is here to cover six fundamental aspects [42] of business processes: What (data description), How (function description), Where (network description), Who (people description), When (time description) and Why (motivation description).

However, business process representations are traditionally built using classical data acquisition methods (interviews, observations, transcription of activities, text analysis, etc.) together with a process representation language such as BPMN or UML. The use of these techniques and specialized languages can easily become hard, complex and time consuming particularly if the organization does not have formal and clear process description documents or if the stakeholders proceed mechanically without really being conscious of the task. On the contrary, participative methods for business process improvement recommend a strong implication of the users [28]. End-users are indeed the ones who have the knowledge and have to use the system in the end, and thus they should really know what is expected. These methods improve the time and quality needed for the acquisition of the useful information to understand and improve the processes. However, the obtained representations are not formalized enough, and they do not correspond to models that conform to a formal modeling language.

Considering these aspects, it is clear that business process modeling cannot been done from scratch by end-users with current modeling languages (too difficult to use): a more user-centered, more soft and friendly approach is necessary to get the foundations of business process models right. To this aim, this article proposes a method which main originality is to take advantage on both BP elicitation participative methods and traditional BP management methods in order to simplify and accelerate the early phases of traditional business process modeling. This method called ISEA [17,34] can be seen as a participative end-user method for business process elicitation and improvement in order to achieve consensual intermediary products that can be mapped in standard languages. Each intermediary product deals with a specific perspective of the business process.

The modeling process of this method is based on participative and ludic technics. It uses a particular domain-specific language that is a pruning and an adaptation of standard languages (particularly BPMN, standard language for business process modeling, and UML, standard language for information systems modeling). Its application conducted by end-users produces intermediary models that can be transformed into analysis models in standard languages that can be used in BPM cycles.

Figure 1 illustrates the two following fundamental characteristics of ISEA:

- *ISEA introduces participative activities to elicit, evaluate and improve business processes in order to simplify the early phases of the traditional BPM life cycle*. Most of BPM methods base their life cycle on the four steps of PDCA (Plan, Do, Check, Act) cycle or adapt it with three main steps: modeling, execution and optimization [13,14,18]. All these life cycles lodge an optimization phase after the implementation. The main goal of ISEA is to reduce the complexity of user requirements phase by facilitating elicitation, evaluation and improvement with participative activities. The intermediary models obtained can be mapped into analysis models conformed to a standard modeling language.
- *ISEA is based on a multi-perspective business process modeling*. ISEA method allows modeling business processes to implement organizational, informational and interactional perspectives. We focused at first on these three perspectives that are particularly suited for modeling and improvement in existing business processes. The organizational perspective tends to identify all the activities performed by functional actors during the process and the documents exchange. The informational perspective is based on the information compiled during the elicitation of the organizational perspective. The interactional perspective, based on organizational and informational perspectives, allows users to create customized interface sketches to test the user interface navigability and the coherence with the process.

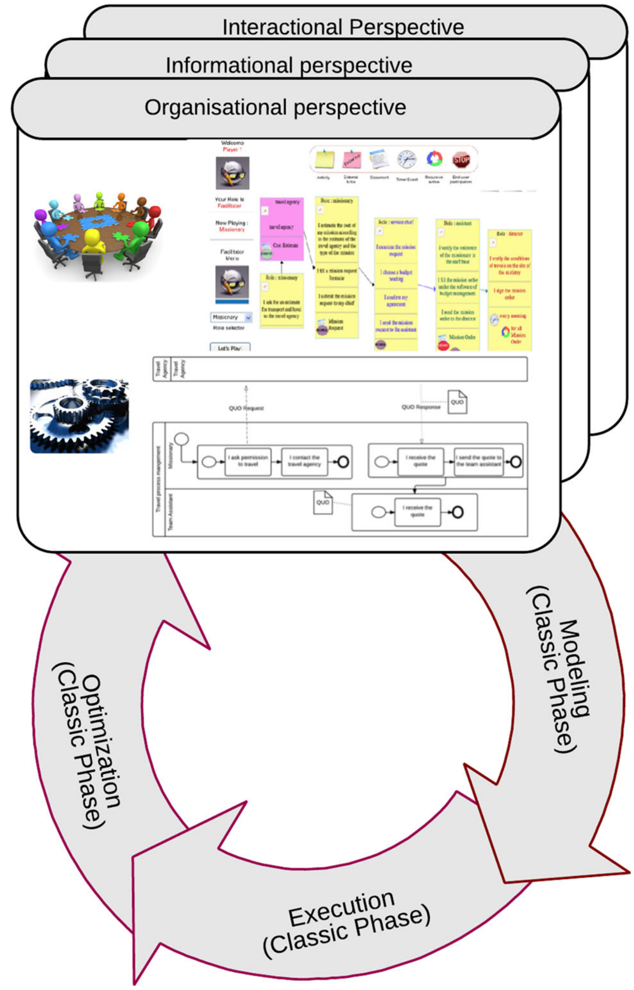


Fig. 1 ISEA life cycle

ISEA is particularly adapted to existing business processes that need to be improved. This kind of improvement, based on participative modeling, is a first step of process improvement that will facilitate, but will not replace the classical optimization phase based on KPI and realized from the process execution. Although it was developed and evaluated in the context of university business processes, it is generic and can be suited to different business process domains to model and improve existing processes.

Section 2 presents related works on participative methods for business process improvement. Then, Sect. 3 gives an overview of the ISEA method: The domain-specific modeling language (meta-model and notation) and the modeling process described by a MAP [15]. Section 4 focuses on a particular path of ISEA, detailing some participative modeling activities proposed in the method. Section 5 presents an experimental research method used to co-construct and validate ISEA, this experimental research method is based on a user-centered experimental validation cycle. Finally, Sect. 6 concludes the paper and describes further works to be carried out.

2 Positioning and related works

Process improvement concerns the set of actions realized to identify, analyze and improve existing business processes to better match with the organizational users' needs.

Many methods for process improvement exist and are emerging as a complement to existing methods for process management. In particular, participative methods are mainly based on quality tools, tend to involve the stakeholders of a process in the proposition of ideas for process improvement, use techniques to stimulate and motivate people and help them to solve problems and to propose creative solutions.

Such methods are based on well-known quality tools such as the seven tools of Total Quality Control [20]: Flow Charts, Ishikawa Diagrams, Checklists, Pareto Charts, Control Charts, Affinity Diagram and Relations Diagram. These tools provide important techniques for self-reflection and analysis. They are simple to use, require no special training and can be used in participative methods to identify innovative solutions to complex problems.

Most of the participative methods are using these quality tools and brainstorming tools to generate new ideas for process improvement. As an example, a cooperative graphic editor was proposed in "CEPE: Cooperative Editor for Processes Elicitation" [32]. This tool allows the specification of processes, the identification of problems and bottlenecks and their requirements for improvement. It provides a space for basic workflow model collective construction, where problems could be discussed, allowing addition of comments and suggestions about each of the workflow stages.

In addition according to the principles of agile modeling such as *assume simplicity*, *embrace change*, *incremental change*, *multiple modeling*, *quality work* and *rapid feedback*, we can count participative methods as a kind of agile modeling [3].

In the same way, the approach called Collaborative, Participative, and Interactive Modeling (CPI Modeling, [6]) focuses on collaboration and participation aspects as being the fundamental of other participative methods. In CPI, Berjis makes emphasis on joint modeling sessions with an active collaboration and participation of the users (business process owners). "It is natural that the end users can provide the most accurate knowledge of their enterprise business processes". The CPI Modeling approach indeed consolidates three aspects:

- *Collaboration* Expert aspect—emphasizes cross-disciplinary collaboration of business analysts, modeling experts and facilitators to organize modeling sessions. The objective is to create a complete enterprise model.
- *Participation* User aspect—emphasizes participation of the business process owners, managers, stakeholders that provide input for modeling. The objective is to provide input information for creating models, validating models and instilling sense of ownership of the models.
- *Interaction* Tool aspect—emphasizes tools for creating models and technologies for enabling collaboration and participation. The objective is to support interaction, facilitate simulation and automate the modeling process.

Based on these considerations, we argue about participative method as an active and end-user method mainly based on quality tools and tends to involve the stakeholders of a process in the proposition of ideas for process improvement [6]. In our opinion, participative method should have a very simplified domain-specific language usable by end-users. Moreover, a business process modeling method should cover multi-perspectives characteristics and be formalized with a well-defined process and language [11]. Depending of this objective, the next section proposes our criteria for evaluating participative methods for business process modeling and improvement.

2.1 Evaluation criteria

The different participative methods can be evaluated by three main criteria that are shown in Fig. 2. These concepts are defined below.

2.1.1 Process

This criterion is related to the modeling process to be followed during the business process modeling and consists into three key factors to evaluate:

- *Conductor profile* defines the profile of the process facilitator who can be an expert modeler or an end-user. Most of the participative methods are indeed conducted by an expert modeler, but we consider as an advantage if the method could be conducted by an end-user.

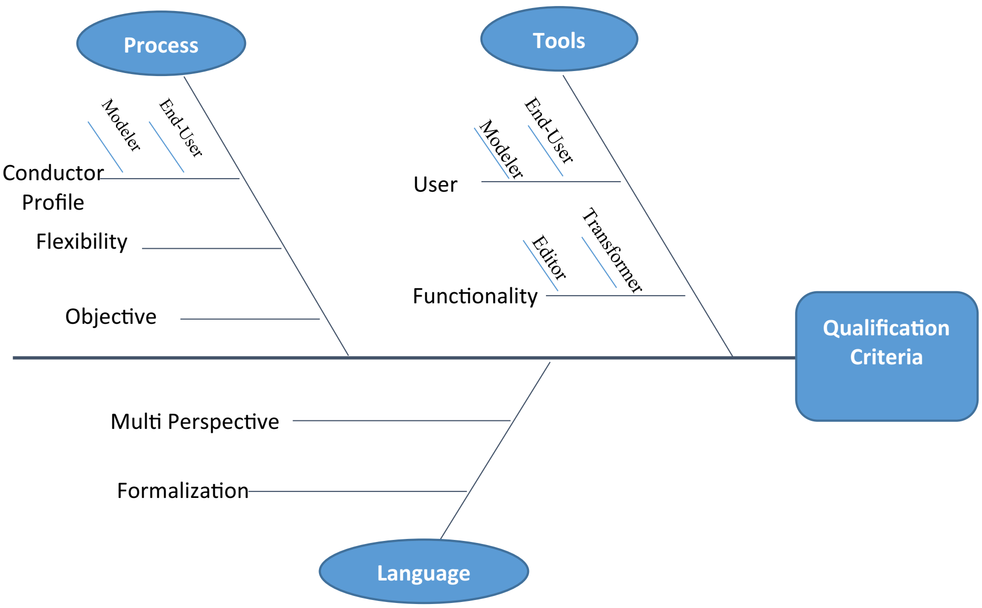


Fig. 2 Method qualification

- *Flexibility* defines adaptability of the modeling process to the project context and needs [2,9,19].
- *Objective* participative methods generally follow two objectives. First of all they try to provide a model of a business process in an organization in order to communicate or to automate the process. Then some of them aim to improve existing challenges in the process.

2.1.2 Language

This criterion is concerned with the language used during the modeling process. This language should be characterized by its formalization level and the multi-perspectives supported.

- *Formalization* one of the important factors in evaluating a language is whether it is formalized or not: is it supported by a meta-model?
- *Multi-perspective* another important factor is the multi-perspectivity of a language. The main following perspectives can be considered:
  - Functional perspective: which process elements (activities) are performed.
  - Behavioral perspective: when and how the activities are performed.
  - Operational perspective: where and by whom in the organization the activities are performed.
  - Informational perspective: which information is manipulated by activities.
  - Goal perspective: why are activities performed.

2.1.3 Tools

Another important criterion for method evaluation concerns the tools supporting the method. In our opinion, such tools have to be softwares. This criterion consists in two factors that are the users of the tool and the functionalities of tools

- *User* who are the users of the tool? Tools are generally targeted to experts, but we consider important in participative methods that tools are also usable by end-users.
- *Functionality* which are the tool's objectives? Objectives of a tool supporting a participative method can be only to create a model, or it can also be to transform the models into executable models.

2.2 Review of participative methods

This section briefly presents some participative method for business process modeling and improvement in order to compare them related to our criteria.

2.2.1 DMAIC

DMAIC (Define-Measure-Analyze-Improve-Control) [30, 38] has been widely used as the method for implementing projects in manufacturing, once its procedures are based on the well-known PDCA (Plan-Do-Check-Act) principles. This method is composed by five stages that should be accomplished in a sequential way for any project:

- *Definition* identification of problems and situations to improve, remarking what is critical to quality;
- *Measurement* selection of critical characteristics to quality through necessary measures in the process;
- *Analysis* identification of the root-causes that are responsible for the identified problems;
- *Improvement* specification of process characteristics to be improved;
- *Control* documentation and follow-up of the new process conditions.

The participative aspects of DMAIC are supported by quality management tools used to improve existing business processes.

2.2.2 PAWS

PAWS (Toward a Participatory Approach to Business Process Reengineering) proposal [7] is to involve the employees of an organization undergoing a business process re-engineering in this task. This involvement should be done in six consecutive phases (Learning, Process Elicitation, Alternatives and Solutions, Option Evaluation, Workflow Implementation and Maintenance) aimed to get acquainted with the method and the objectives of the project, identify problems of the current processes, generate optional re-engineering solutions, generate the proposed process model and finally validate the proposed model. [8]

2.2.3 EKD: 4EM

EKD—Enterprise Knowledge Development method [10] is a representative of the Scandinavian strand of EM (Enterprise Modeling) methods. It defines the modeling process as a set of guidelines for participative way of working and the modeling product in terms of six submodels: Goals model, Business Rules Model, Concepts Model, Business Process Model, Actors and Resources Models, Technical Factor and Requirements Model. Each model focuses on a specific aspect of an organization.

The ability to trace decisions, factors and other aspects throughout the enterprise is dependent on the use and understanding of the relationships between the different submodels addressing the issues. When developing a full enterprise model, these relationships between factors of the different submodels play an essential role [36].

The method "For Enterprise Modeling Method (4EM)" is an evolution of EKD and consists of three central elements that can be seen as fundamental concepts. Each of these elements is strongly linked to the other elements, but nevertheless they can be into:

- A defined procedure for the modeling, where the notation is determined,
- Conduction of the modeling in form of a project with defined roles,
- A participative mode of practice [36].

2.2.4 C3S3P

C3S3P (Concept Study, Scaffolding, Scoping, and Requirements Modeling) is based on works in several European projects from the area of networked and extended enterprises. An extended enterprise is a dynamic networked organization, which is created ad hoc to reach a certain objective using the resources of the participating cooperating enterprises. In order to support solutions development for extended enterprises, the external project [22] developed a methodology for extended enterprise modeling [23], which initially was named SGAMSIDOER based on the abbreviations of the different modeling steps proposed: Scoping of the extended enterprise, Gather existing partner information, Analyze extended enterprise potential, Model extended enterprise, Simulate and analyze model scenarios, Implement model, Deploy extended enterprise, Operate, as well as Evaluate and Reengineer extended enterprise. C3S3P, like SGAMSIDOER, aims at executable solutions based on visual enterprise modeling (EM). The seven C3S3P phases are:

- *Concept testing* pre-studies are performed to investigate whether EM is a suitable and accepted way of developing executable solutions for the networked enterprise.
- *Scaffolding* aims at creating shared knowledge and understanding among the participants of the project about the scope and challenges of the project.
- *Scenario modeling* creation of executable models supporting the networked enterprise in the defined scope including all relevant dimensions required, like process, product, organization or IT-systems.
- *Solutions modeling* refining the scenario model by integration personnel, product structures, document templates and IT systems required for using the enterprise model in an actual project.
- *Platform configuration* configure the solution models for use in the networked or extended enterprise by connecting the enterprise model to the platform used.
- *Platform delivery* encompasses the roll-out of model-configured solutions.
- *Performance improvement* captures indicators for process and product quality and uses adequate management instruments.

During the deployment all the mentioned methods, it is necessary to have a modeling expert or a business analyst as a facilitator to organize modeling sessions.

2.3 Synthesis

Table 1 synthesizes the different participative methods studied in previous section in relation to our criteria.

In general, participative methods are not based on standard modeling languages and are not integrated in a traditional business process development cycle. Few of them propose a multi-perspective modeling.

Processes of participative methods are generally conducted by an expert, and they are not flexible to adapt to different types of projects. Some of these methods are developed only for automatizing the existing business process and not for deploying an improvement of this process.

Moreover, the tools that support these methods are not end-users oriented and need to be conducted by an expert modeler during the business process modeling. In addition, the result of these tools should be synthesized by an analyst and they do not have any possibility to transform their results to standard languages.

In order to overcome these drawbacks, we propose in the next sections a participative end-user method for multi-perspective business process elicitation and improvement called ISEA. ISEA is based on the traditional BPM life cycle and combines it with a participative approach, based on quality tools and role games. In addition, ISEA supplies end-users with a language and participative activities allowing to elicit and improve complementary perspectives of business process.

Table 1 Synthesis of participative methods

Methods	Method qualification													
	Process						Tools				Language			
	Conductor profile		Flexibility		Objective		User		Functionality		Multi-perspective		Formalized	
	Modeler	End User	Yes	No	Automatization	Process improvement	Modeler	End User	Editor	Transform	Yes	No	Yes	No
DMAIC	✓	–	–	✓	–	✓	✓	–	✓	–	–	✓	–	✓
PAWS	✓	–		✓	✓	✓	✓	–	✓	–		✓	–	✓
EKD	✓	–	✓	–	✓	✓	✓	–	✓	–	✓	–	–	✓
4EM	✓	–	✓	–	✓	✓	✓	–	✓	–	✓	–	✓	
C3S3P	✓	–	✓	–	✓	✓	✓	–	✓	–	✓	–	–	✓

3 ISEA: a participative end-user modeling method for business process

ISEA is defined by a modeling process, a modeling language and a tool. Sections 3.1 and 3.2, respectively, describe the language and the process. The tool ISEAsy supporting the method is also illustrated. Use of the process, language and tool will be illustrated through a partial example of the mission management process in a university, on the scenario of a missionary aiming to travel in France for a conference.

3.1 Modeling language

This section introduces the domain-specific language dedicated to the different business process perspectives. This DSL is very simple and uses graphical notations such as post-its or trees in order to make them understandable and usable by functional actors in individual and collective modeling activities (such DSL are used in similar works like PICTURE [5]). This DSL is defined by a meta-model defining its abstract syntax, whereas its notation defines the concrete syntax of the language elements.

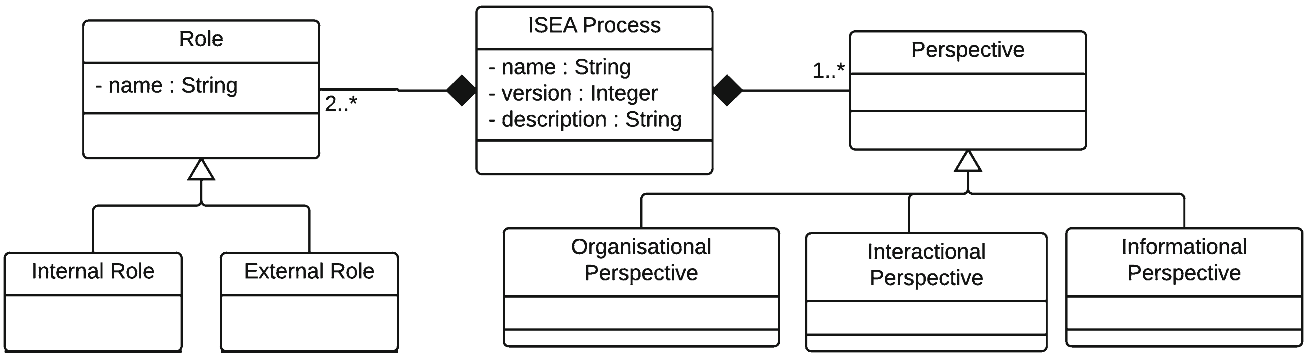


Fig. 3 Global view of the meta-model

3.1.1 Global view of the DSL meta-model

An ISEA process includes three perspectives (see Fig. 3): organizational perspective, informational perspective and interactional perspective. Moreover an ISEA process implies several roles that can be either internal roles or external roles. An ISEA process is defined by a name and a description, and may be versioned in case of evaluation and improvement. In ISEA, versioning a process means duplicating it and making it evolve.

Meta-models of organizational and informational perspectives are described below. The meta-model of the informational perspective is under construction and will be part of a future publication.

3.1.2 Organizational perspective

The organizational perspective (see Fig. 4) is seen as the representation of the interventions of all actors of the process. More precisely, an intervention can be either an external intervention of an external role or an activity of an internal role. Interventions can be linked together through flows that can be in input or output of others activities. As versioning is considered as a duplication of the process, an intervention is attached to only one organizational perspective.

Activities (see examples in Fig. 5) are composed of several actions sorted in different types:

- Individual action: describes what an internal roles do in the process; usually begins with an "I," for example "I estimate the cost of the mission…"
- Create document: if an activity requires a new document, the participant uses the "Create Document" action to create an instance of the Document class. For example, a document "Mission Request" is created by the missionary in the action "I fill a mission request formular."
- Reuse document: if an activity requires reading or updating a document instance that was beforehand created in the process, the participant uses the action "Use document." For example, the document "Cost Estimate" created by the travel agency (external role) is reused by the missionary during the action "I estimate the cost of the mission according to the estimate of the cost…."
- Recursive actions: indicates that the activity is iterative. For example the director checks and signs all the mission orders.
- Temporal action: represents a temporal event in the activity. For example, the director checks and signs the mission orders every morning.
- Stop action: indicates that the participant considers his intervention in the process as finished. For example, once the mission orders checked and signed, the director does not interact any more in the process.

An acronym, a description and eventually a pdf file characterize the different documents needed in the process.

Therefore, an organizational perspective is mainly described by the interventions (external interventions and activities) of external and internal roles. Moreover, an organizational perspective is also described by individual difficulties identified by internal roles. Similar individual difficulties are grouped into consensual collective difficulties. In order to resolve these collective difficulties, individual improvement actions can be expressed by internal roles. The individual improvement actions are also grouped into collective improvement actions that will allow the process to be improved and replayed. Sometimes, a collective improvement actions may help to achieve others collective improvement actions, making into evidence a form or prioritization of the collective improvement actions. We will see in Sect. 3.2 that difficulties and improvements actions are elicited during the evaluation of the organizational perspective.

Figure 5 illustrates the concrete syntax of external interventions and activities of the organizational perspective. The notation for difficulties and ameliorations will be illustrated in Sect. 4 (see Fig. 13).

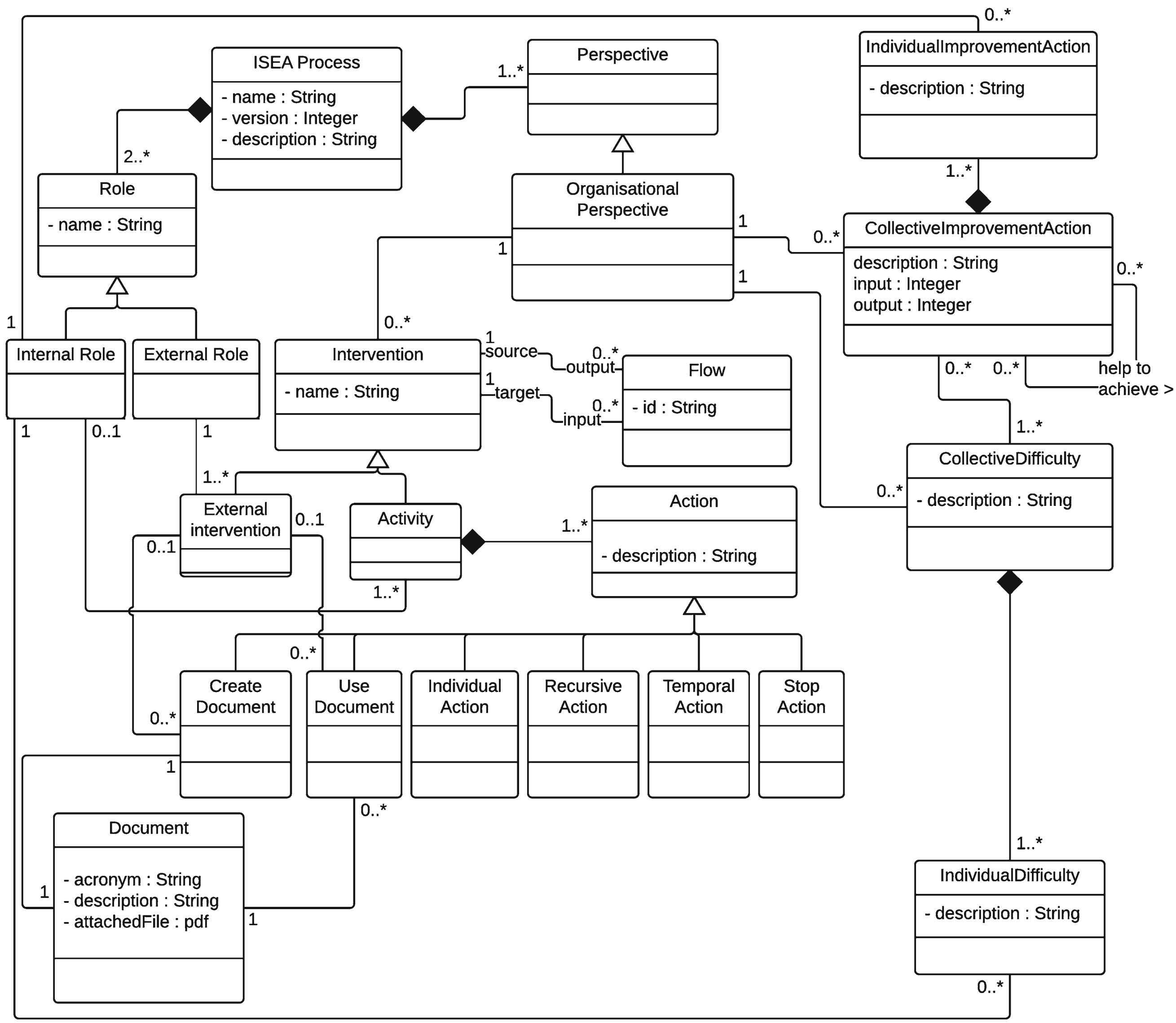


Fig. 4 Meta-model of the organizational perspective language

3.1.3 Informational perspective

The meta-model of the informational perspective is represented in Fig. 6. The informational perspective uses documents (imported from the organizational perspective) in order to construct fragments of documents corresponding to one or more information (e.g., "name," "address," …) that will then be linked to one concept (e.g., concept "Missionary"). The goal of the informational perspective is indeed to identify the different concepts belonging to the ISEA process. Concepts can be hierarchized into subconcepts.

As for interventions in the organizational perspective, documents and concepts are considered to belong to only one informational perspective due to the duplication of the processes in case of versioning.

Figure 7 partially illustrates the concrete syntax of the informational perspective language. A tree represents an informational perspective; each concept is represented by a branch and each information by a leaf. The association between a fragment and its information is maintained by hyperlinks in the tool (not represented in Fig. 7).

3.2 ISEA modeling process

Figure 8 describes by two maps [15] the ISEA modeling process. The global ISEA map has two intentions in addition to Start and Stop: *Elicit intermediary models* and *Construct analysis models*. The different sections of this map are described below.

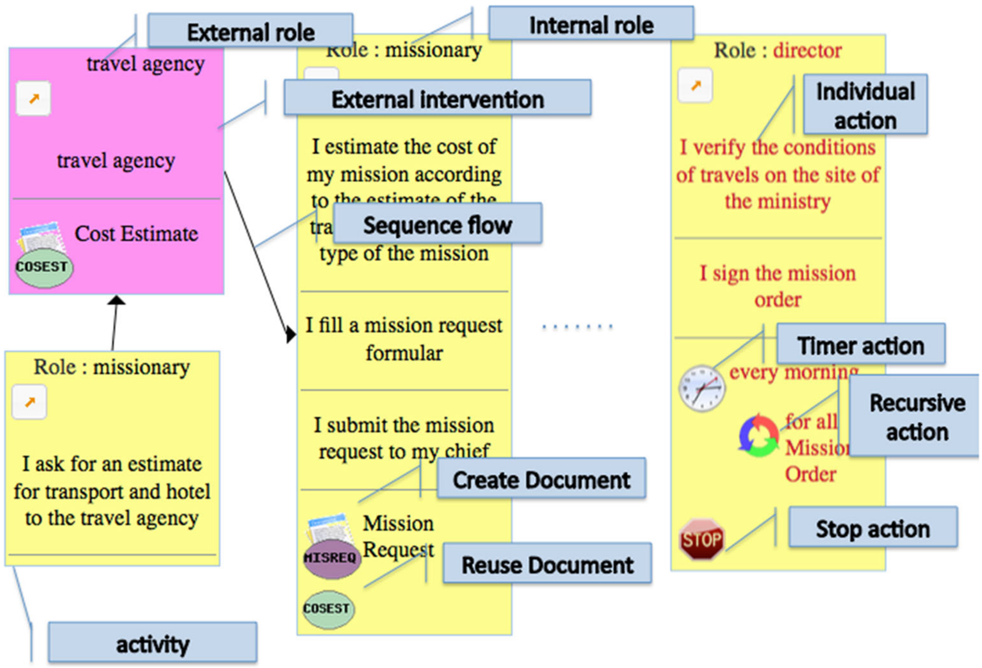


Fig. 5 Concrete syntax of interventions of the organizational perspective language

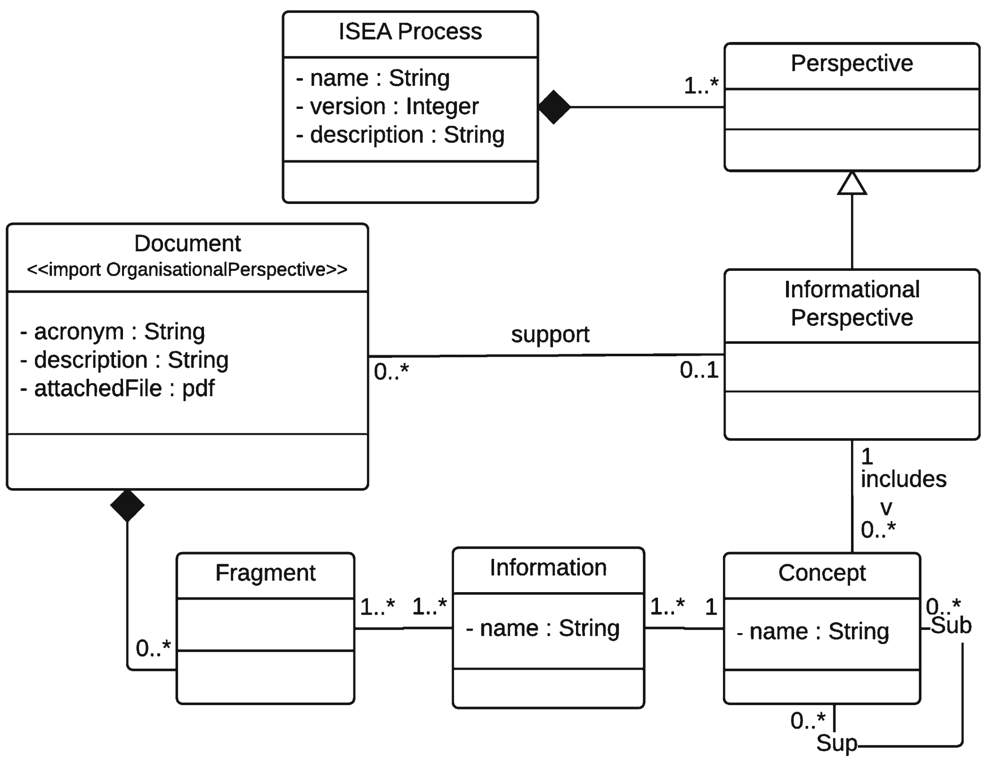


Fig. 6 Meta-model of the informational perspective language

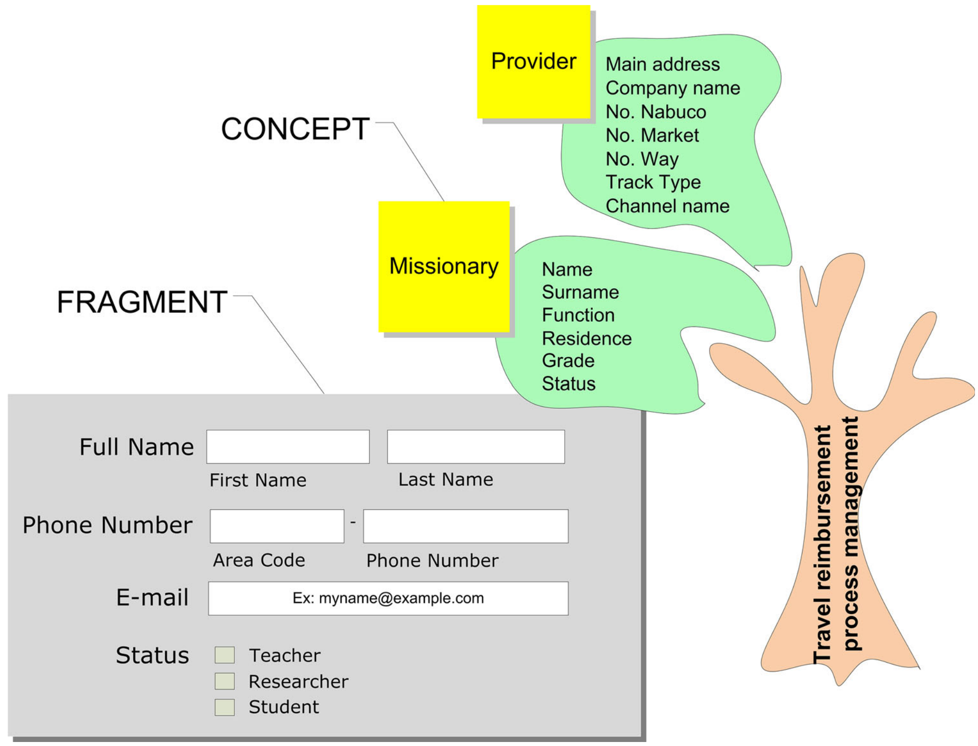


Fig. 7 Concrete syntax of the informational perspective language

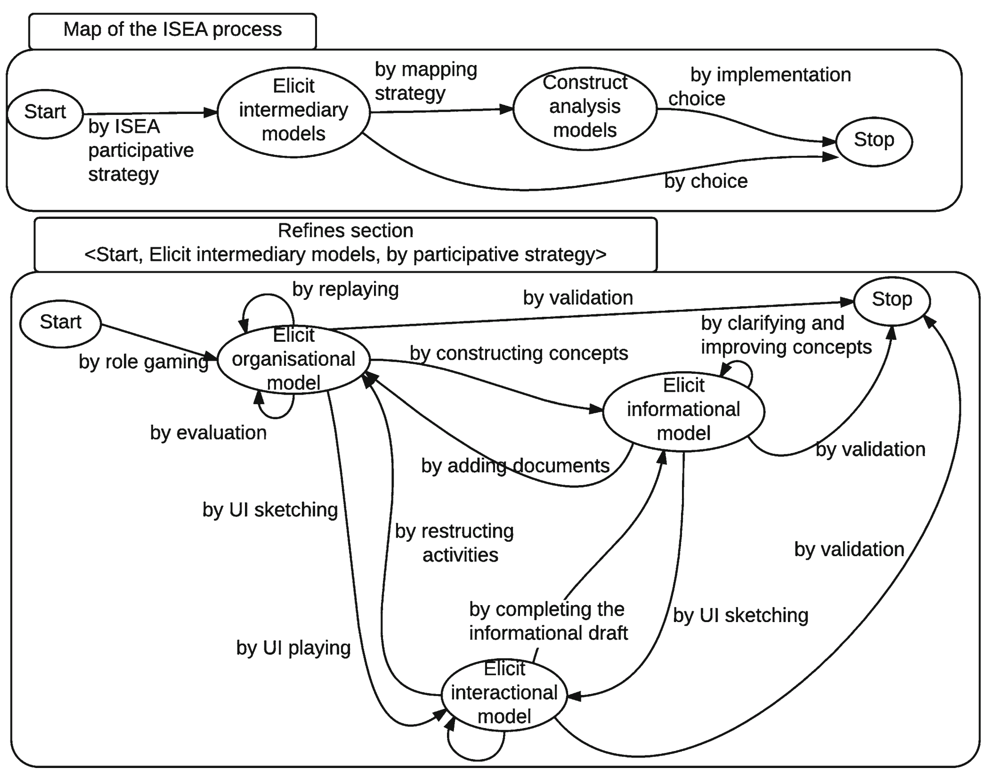


Fig. 8 The ISEA modeling process described as a map

- ⟨*Start, Elicit intermediary models, by ISEA participative strategy*⟩. This section represents all the participative and playful activities realized by the functional actors to obtain, evaluate and improve intermediary models of the business process. An intermediary model represents a process perspective described in the domain-specific language (DSL) proposed in Sect. 3.1. This section is the core of ISEA and will be detailed further.
- ⟨*Elicit intermediary models, Stop, by choice*⟩. If the goal of business process cartography is to facilitate the communication in the business team by a better understanding of each actor's role, the first intention (elicitation) is sufficient. In this case, the elicitation of the business process representations is useful to communicate, evaluate and improve existing processes within organizations.
- ⟨*Elicit intermediary models, Construct analysis models, by mapping strategy*⟩. This section represents the mappings between intermediary models and standard models (BPMN, Entity-Association, UI models). These transformations are necessary if the final goal is to automate the business process (see section ⟨*Construct analysis models, Stop, by implementation choice*⟩) but thus to benefit from functionalities of CASE tools. In the contrary to the activities proposed in the previous section, such transformations are entirely automated. For now, only the transformation of the organizational perspectives into BPMN models is implemented. Each concept of the organizational meta-model maps with a BPMN collaboration diagram concept. For example, an *external role* is represented by a *black box* and an *ISEA process* is represented by a BPMN *white box*. An *internal role* is a *lane of the white box*. An *activity* is represented by a *BPMN subprocess*, the *flows between external roles and internal roles* are represented by *message flows* and *flows between two activities* are represented by *sequence flows*. This transformation step will not be more detailed here, all the mapping rules can be found in [33].
- ⟨*Construct analysis models, Stop, by implementation choice*⟩. The aim is to automate the process. In this case, the intermediary models (transformed into analysis models) constitute consensual requirements allowing the development of process-aware information systems. An analyst will probably have to enrich these analysis models in order to get computerized models.

The refined map refines the section ⟨*Start, Elicit intermediary models, by ISEA participative strategy*⟩. It contains three intentions that correspond to each perspective: *Elicit organizational model, Elicit informational model* and *Elicit interactional model*. All the proposed strategies are based on participative and playful methods. Each strategy is supported by a partially ordered set of individual or collective activities. For example, the strategy *by role gaming* is supported by several activities mainly collaborative: the functional actors of a business process, with the help of an animator, collectively elaborate the description of the business activities and the exchanged documents.

Generally, a strategy between two intentions allows *eliciting* a perspective from one or several others. For example, the informational perspective is in general elicited using the organizational perspective; the interactional perspective needs the two previous ones (path: ⟨*Start, Elicit organizational model, by role gaming*⟩, ⟨*Elicit organizational model, Elicit informational model, by constructing concepts*⟩, ⟨*Elicit informational model, Elicit interactional model, by UI sketching*⟩).

A strategy between two intentions can also *improve* a perspective from another one. For example, if the informational perspective contains data that are not in the documents invoked in the organizational perspective, the section ⟨*Elicit informational model, Elicit organizational model, by adding documents*⟩ allows completing the organizational perspective.

Recursive strategies are dedicated to evaluate and/or improve a perspective. For example: ⟨*Elicit organizational model, Elicit organizational model, by evaluation*⟩ highlights the process difficulties and the possible improvements of the organizational perspective.

As an illustration, a particular path of the map is detailed in the following section.

4 Exploring a path in ISEA

This section illustrates different steps followed during process modeling in ISEA. Figure 9 describes a path of the ISEA process. This path allows:

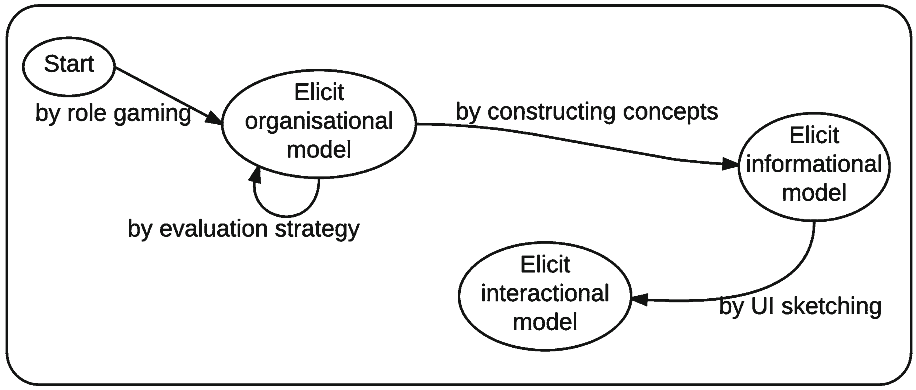
- to elicit the organizational perspective through a role-playing game in which each participant plays its own role,
- to evaluate this perspective by a strategy dedicated to identify the difficulties and possible improvements,
- to elicit the concepts of the informational perspective by the exploitation of the documents identified in the organizational perspective,
- to elicit the interactional perspective via the sketching of the user interactions during their business activities which have been identified in the organizational perspective.

Individuals and collectives activities are illustrated in next sections using ISEAsy, the tool support of the method ISEA.

4.1 ⟨Start, Elicit organizational model, by role gaming⟩

The goal here is to elicit an organizational model corresponding to a business process expressed using a very simple DSL

Fig. 9 A path of the ISEA process



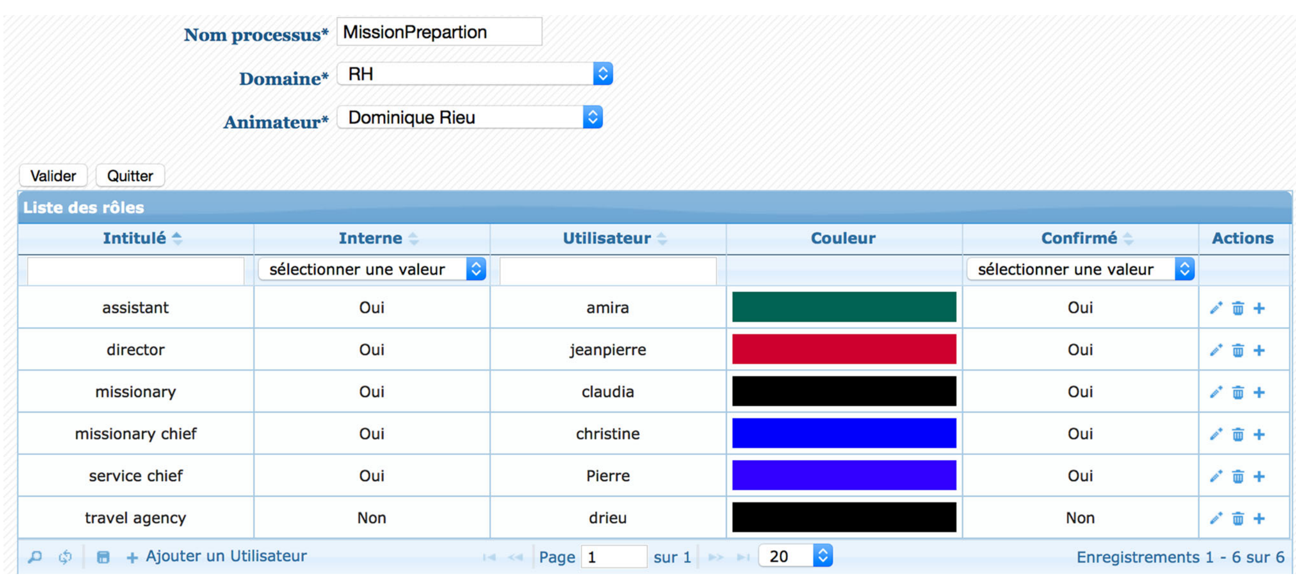
and representing all the activities and document exchanges (see Sect. 3.1). In the role-playing game as in the whole ISEA method, all stakeholders are involved, and more particularly the end-users, who are the domain experts and possess the necessary knowledge of how the processes should operate, which tasks have to be carried out, which business rules need to be enforced, which validation checks need to be performed, etc.

An elicitation session begins with the **identification** of the internal roles of the process and of the functional actors which can play these roles. Identification of actors consists in contacting the first person in the process (often the initiator of the process) who will quickly identify the people with whom he interacts, and so on. During the identification activity, the external actors can be identified, but this identification can be incomplete and external actors can appear during the game.

After the identification activity, all the functional actors are gathered. The session is carried out in two hours and is leaded by a facilitator. Before playing, the session begins with the following activities:

- *Reception* the facilitator describes the general context of the session through electronic slides.
- *Roundtable and role assignment* the facilitator presents the different roles of the process and assigns each role to a participant. Generally each participant plays the same role as he has in real life. In the tool support ISEAsy, a role is depicted by a color (see Fig. 10). During the role assignment, it may be that some roles have not been identified. In this case, the facilitator can play this role or a new session will be necessary. At the end of this activity each participant has a role. Moreover, the facilitator represents the external actors.
- *Scenario proposal* a scenario must be collectively discussed. It must be precise and limited enough to be played in less than two hours. For example "A missionary has a conference in France and he has to travel."

After the previous activities (less than 20 min), the game can start. Figure 11 shows the participants playing with the role-playing game in ISEAsy. Each participant assumes a role


Nom processus* MissionPrepartion
Domaine* RH
Animateur* Dominique Rieu
Valider Quitter
Liste des rôles

Intitulé	Interne	Utilisateur	Couleur	Confirmé	Actions
	sélectionner une valeur			sélectionner une valeur	
assistant	Oui	amira		Oui	
director	Oui	jeanpierre		Oui	
missionary	Oui	claudia		Oui	
missionary chief	Oui	christine		Oui	
service chief	Oui	Pierre		Oui	
travel agency	Non	drieu		Non	

+ Ajouter un Utilisateur Page 1 sur 1 20 Enregistrements 1 - 6 sur 6

Fig. 10 Role assignment

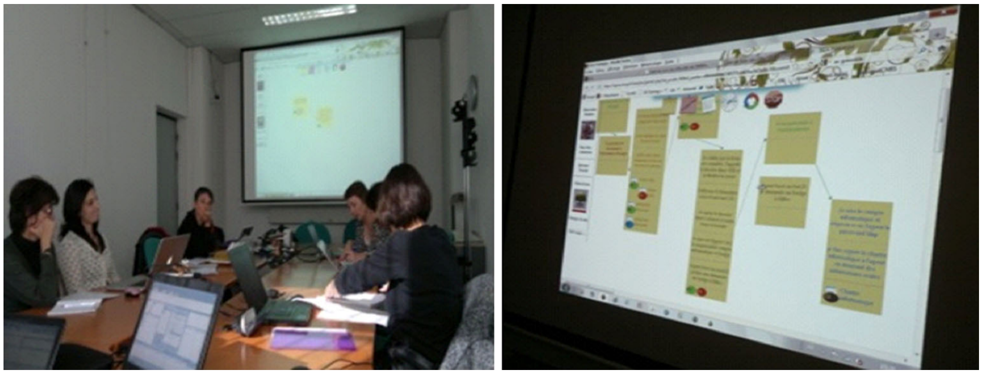


Fig. 11 The role-playing game using the tool ISEAsy

and acts out a real-life situation in order to get in a participative way, a description of the daily activities encountered during a specific process. Participants take their turn, one after the other, depending on the situation, as would occur in real life. As an example, in a travel preparation process, the game begins with the missionary, who needs to estimate the cost of a mission.

A participant places a yellow post-it on the workspace to represent an activity he accomplishes during the process. He uses a set of graphical elements (see Sect. 3.1) with which he represents the actions performed during the activity. If the intervention of an external actor is necessary, a pink post-it is added, where no action is noted on and only documents may move on.

When the participant has finished to describe an activity, he draws one or more arrows handing over the turn to the next participants.

4.1.1 Results

Figure 12 shows the result of the role-playing game in the tool ISEAsy, support of the ISEA method. The result is very similar to a BPMN basic process. Just like the models described in BPMN, the organizational perspective includes behavioral (activities) and informational (documents) dimensions. The links between activities have the color of the internal role that draws the link. During the game, the activities and links can be deleted or modified only by the participant who creates them.

4.2 ⟨Elicit organizational model, Elicit organizational model, by evaluation strategy⟩

In traditional BPM methods, an optimization phase is usually performed once the modeling and execution phases have been completed, in order to measure and monitor the current performance gaps between current and desired performance of the process. On the contrary, the evaluation strategy (section ⟨*Elicit organizational model, Elicit organizational model, by evaluation strategy*⟩) allows early detection of difficulties encountered by the participants during the process, to make it possible to improve the process before implementation. The goal is to detect the difficulties encountered during the process execution and thus propose potential improvement actions. This strategy is supported by quality improvement tools: brainstormings, affinity diagrams and relation diagrams are used to identify, organize and prioritize the improvement actions. The strategy is composed of several individual and collective activities described below:

- *Identify difficulties* The business process representation produced during the role-playing game is discussed with the purpose of identifying and resolving inconsistencies between the process model and the participants' experience in the business process. The goal is to identify the difficulties associated with the business process being analyzed. Difficulties of the Mission Management process could be for example "Insufficient number of personnel" or ""Procedure not yet established" (see Fig. 13). During an individual silent brainstorming, participants are invited to indicate any difficulty they have experienced in the business process. Each participant marks different difficulties associated with the process or activities that take place during the business process. Each difficulty is characterized by the role of the participant and a brief description. Participants are then asked to exchange ideas and to discuss the difficulties associated with the business process analyzed.
- *Sort difficulties into related groups* In this step, the facilitator proposes to sort the ideas into groups related to reviewing and analysis. Participants discuss the difficulties encountered in the business process and group them by similarity using an affinity diagram. An affinity diagram (see Fig. 13) is a quality management tool used to organize ideas and data. It is commonly used within project management and allows large numbers of ideas

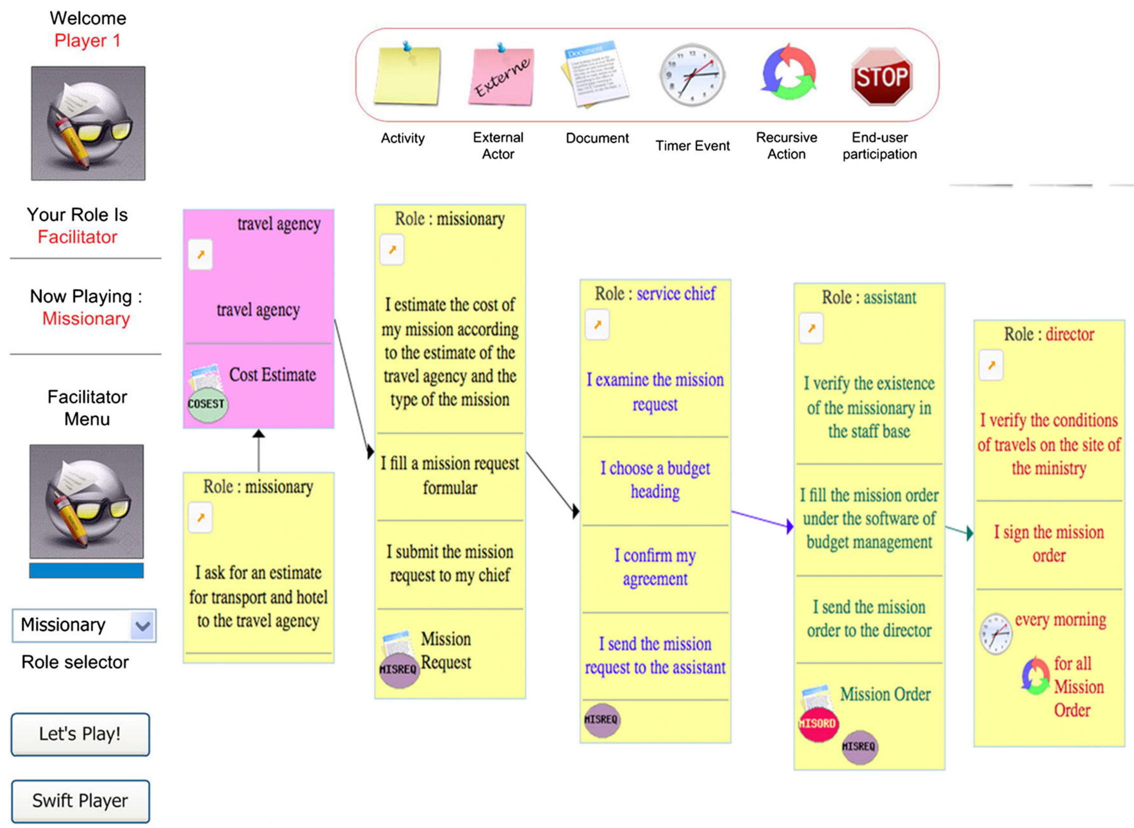


Fig. 12 Organizational perspective: a business process representation in ISEAsy

stemming from brainstorming to be sorted into groups for review and analysis [10]. If a participant disagrees with the placing of an idea in a certain group, he or she can move it. The game indeed creates an environment in which it is acceptable to disagree with people having a different viewpoint. If a consensus cannot be reached, the facilitator duplicates the idea and places one copy in each group. The facilitator then proposes a header idea that captures the essential link among the ideas contained in each group. This heading consists of a phrase or sentence that clearly conveys the meaning of the difficulty. A group of difficulties is subsequently called "Consensual difficulty."

- *Identify improvement actions* Once the difficulties have been identified, the participants use silent brainstorming to generate and prioritize opportunities for improvement (called improvement actions). Examples of improvement actions could be for example "Establish clear procedures" or "Identify clearly the roles of each person". The brainstorming question is: *"Which actions are required to solve the difficulties?"* Each participant notes down on virtual post-its the possible improvement actions, listing only one idea per post-it.
- *Sort improvement actions into related groups* In the same way as for the difficulties, the participants sort the improvement actions into related groups using an affinity diagram. Once the participants have completed the exercise, the facilitator selects a heading for each group. A group of improvement actions is subsequently called "consensual improvement action".
- *Identify priorities for improvement actions* The goal is to determine priorities between the different consensual improvement actions using a relations diagram (see Fig. 14). Here, the participants are asked to answer to the following question for each action: *"Is the implementation of this action useful to realize any other action?"* The facilitator draws arrows from each action to the ones it influences and repeats the question for each action. When all the relations between the consensual improvement actions are established, the relations diagram is completed by the arrows in (I) and out (O) for each idea. The number of arrows determines the order of execution of improvement actions. The ones with the largest quantity of arrows are the key ideas.

4.2.1 Results

The results of the evaluation strategy complete the organizational model by a consensual set of difficulties related to the process and a list of required improvement actions

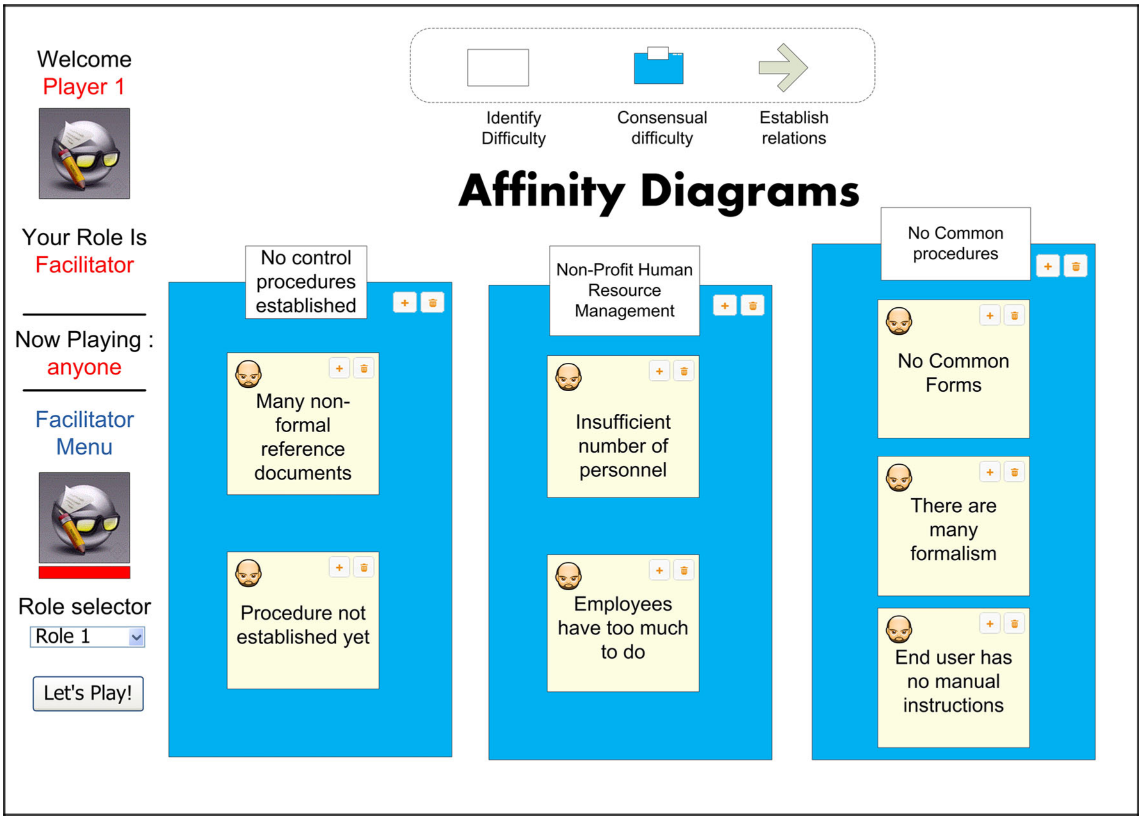


Fig. 13 Affinity diagram in ISEAsy

with a priority order. More precisely, an affinity diagram (see Fig. 13) shows the difficulties and a relations diagram (see Fig. 14) shows the order of priority in the list of improvement actions.

The strategy "by replaying" (see Fig. 8) could then be raised in order to "replay" the process according to the proposed improvement actions. According to the order of execution of the improvement actions, the participants would update their activities by imagining performing the improvement action and replay the process. Activities of the original process may be added, modified or deleted. Several iterations may be needed until all actions have been processed or until a suitable satisfaction level on the business process is reached. The result of this strategy would be a representation of an optimal business process, where several improvement actions are proposed and would be achieved with a satisfying satisfaction degree.

4.3 ⟨Elicit organizational model, Elicit informational model, by constructing concepts⟩

Goal is to elicit the concepts (information) of the informational perspective and their attributes by a fragmentation of the documents elicited in the organizational perspective. Three main individual and collective activities are proposed:

- *Cutting* In a first phase, the facilitator gives 2 or 3 documents to each actor who is individually asked to cut in each document the different fragments that seem pertinent to be grouped together. For example (see Fig. 15), a participant may cut fragments on a document corresponding to information on a missionary (name, sex, service, etc.) and in the same document, information on the travel (departure date, arrival date, etc.). This activity is individual and must not exceed ten minutes; otherwise, the participants may be bored.
- *Model elaboration* Participants are collectively asked to place the document fragments on a tree symbolizing a tree of concepts. One after the other, the actors place the different fragments either in a new branch of the tree symbolizing a new concept (for example, the new concept Travel), or in an existing branch symbolizing new elements to describe an existing concept (for example, information added to the concept Missionary) (see Fig. 16). When all fragments are placed on the tree, the individual activity "Cutting" iterates until the whole documents are cut.

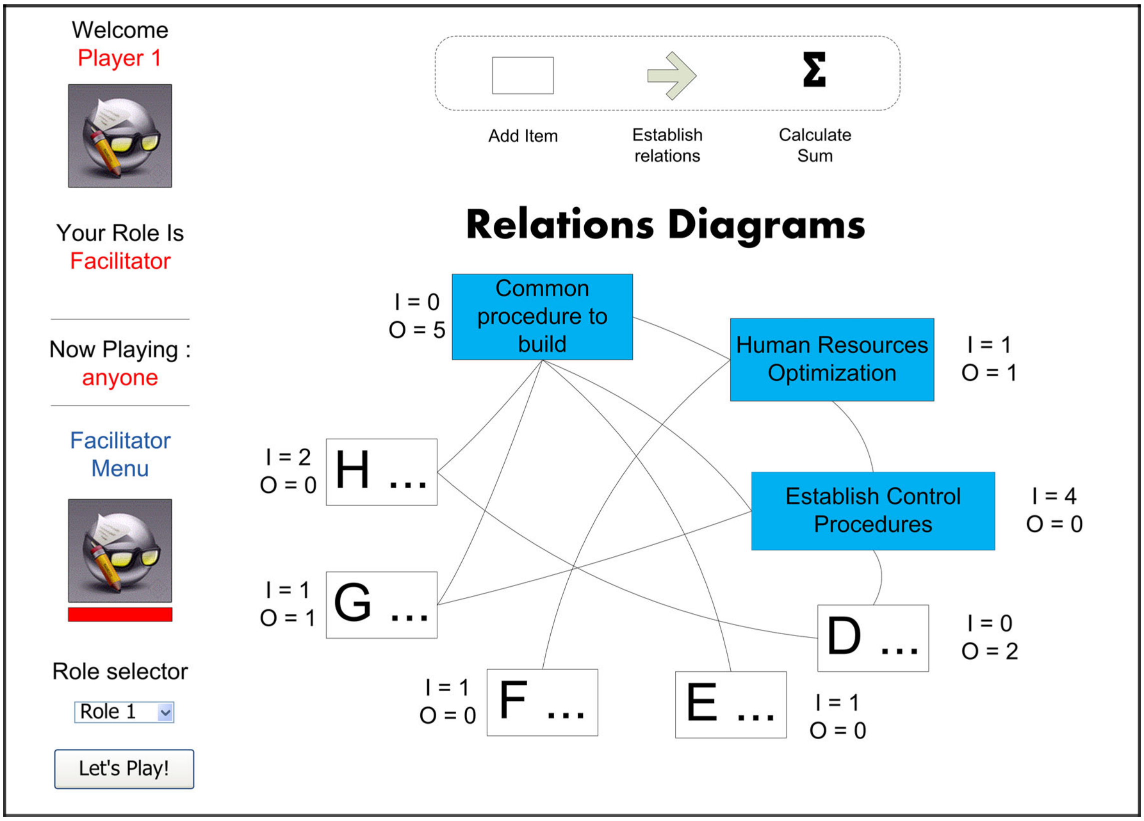


Fig. 14 Relations diagram in ISEAsy

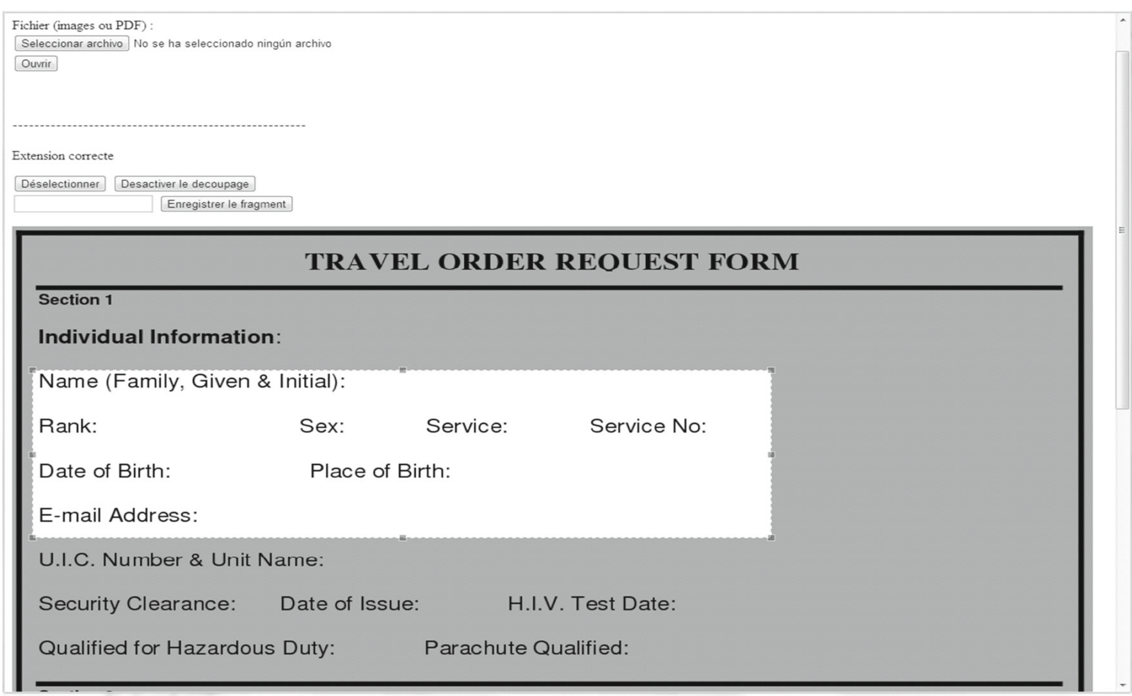


Fig. 15 The “cutting” activity: a fragment in a document

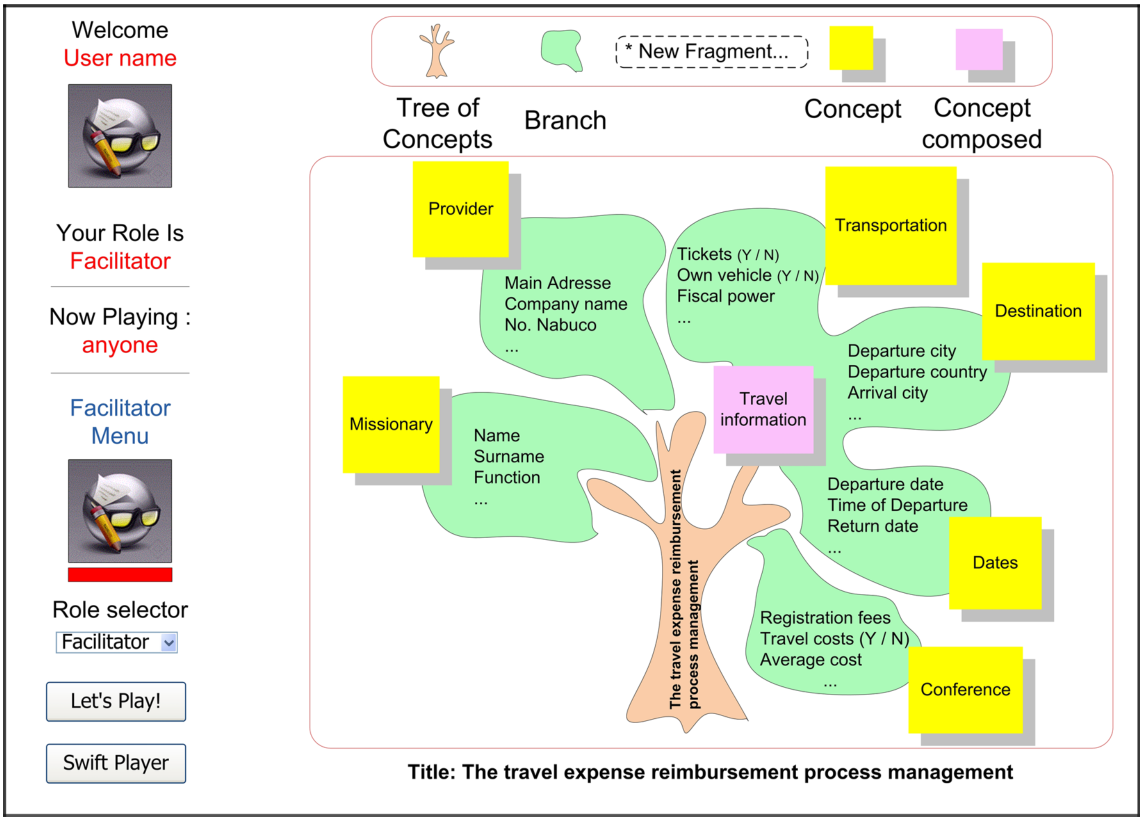


Fig. 16 Informational perspective: the tree of concepts in ISEAsy

4.3.1 Conflict resolution

Several fragments of different documents may represent the same information. For example, name and date of birth of a missionary may exist on different documents. In this case, actors place the information on the same branch, and the facilitator will take a time to resolve these conflicts.

4.3.2 Results

The result is a tree of concepts (see Fig. 16) representing a simple domain model conformed to the meta-model described in Sect. 3.2: each branch corresponds to a concept or a subconcept, a leaf represents an information.

4.4 ⟨Elicit informational model, Elicit interactional model, by UI sketching⟩

The goal is to create customized interface sketches to test the user interface navigability and its coherence within the business process. As for the other perspectives, different individual and collective activities are proposed:

- *Sketching* From the business process model resulting from the organizational perspective elicitation and the tree of concepts resulting from the informational perspective elicitation, each participant is asked to imagine the user interface he would like in order to realize his activities in the most efficient way, and to resolve the potential difficulties identified during the organizational model evaluation. For example, if a consensual difficulty is expressed during the organizational perspective evaluation concerning the too important number of message exchanges between the missionary and his chief to initialize the mission process, the missionary may first imagine a menu where he could estimate the price of the mission, look for an estimation of the price of the hotel and transport on adequate web sites, look for the conference rates on the conference website, and only then contact his chief to get an approval (screen of the missionary in Fig. 17). In the same way, the chief could imagine the sketching of the user interface allowing him to receive emails when he has to validate or refuse a request (screen of the team leader in Fig. 17). To construct their interfaces, participants can use existing UI sketching tools such as Balsamiq [4] (Balsamiq, s.d.)

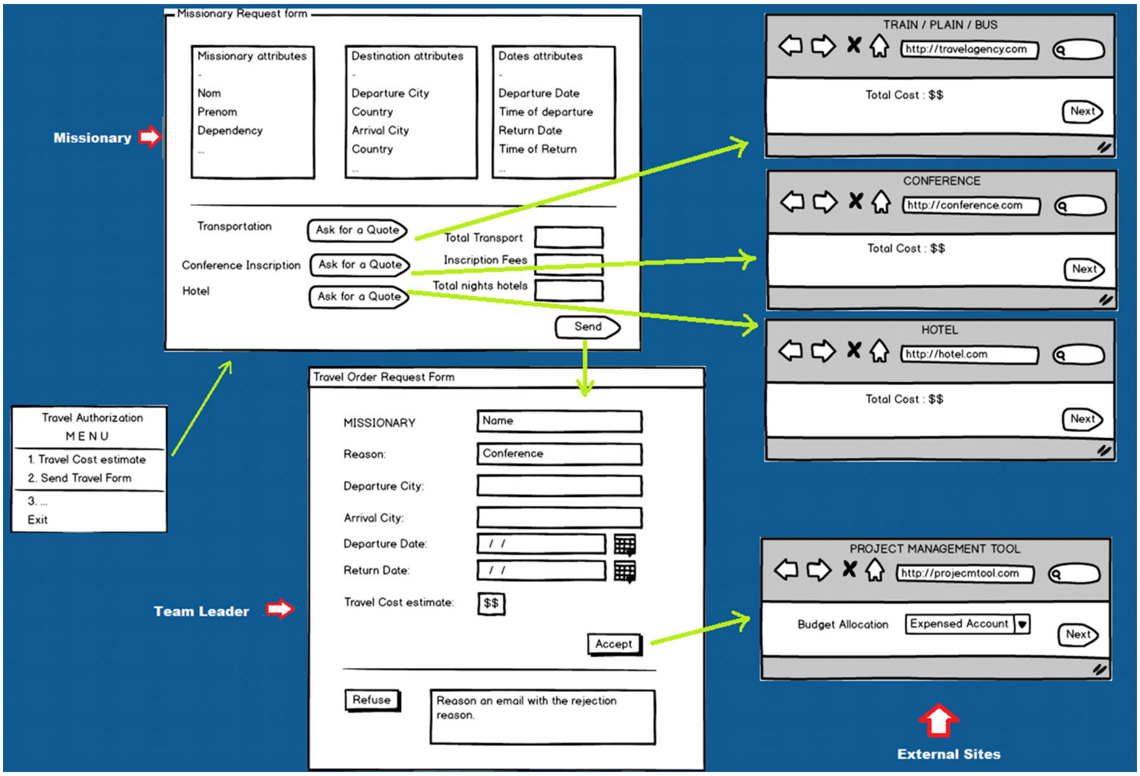


Fig. 17 Interactional perspective: navigation between sketches

- *Navigation validation* The facilitator draws the navigation between the proposed sketches using tools such as Gambit [31]. Participants are then asked to validate the navigation or to correct their interfaces in order to be satisfied.

4.4.1 Results

The result of the interactional model elicitation is a representation of the ideal interfaces and screens navigations between actors (see Fig. 17). The proposed interfaces could then be transformed and completed by a user interface specialist in executable and standardized UI models.

This ideal interactional model is a representation of an improved business process, which may not be coherent with the other perspectives, in particular with the organizational perspective. In this case, the strategy *by restructuring activities* (see Fig. 8) should be executed in order to align the two perspectives.

5 Validation of the ISEA method

The modeling language and the modeling process of the method ISEA were designed and validated adopting a user-oriented method validation cycle that we proposed as a previous research result [25] for the co-construction and validation of a method. This user-oriented validation cycle (see Fig. 18) is based on the integration of user-centered experimental practices and is composed on three stages:

- Analysis aims to validate the language dictionary and to sketch the language notation (concrete syntax) and the modeling process.
- Design allows to validate the language notation with criteria such as the pragmatic quality or the complexity of a model [26],
- Operationalization is dedicated to validate language and process support tools.

During analysis, the methods are validated by a restrict circle of users, on average two business processes, each business process involving from six to eight actors. This restrict circle is spread during design (from 4 to 8 business processes). During operationalization, the tools are generally validated by the same users as in design so as to measure the acceptability degree of the method in a mediatized mode. During each stage (analysis, design, operationalization), experiments are led in a purpose of validation, but also exploration and co-construction with the future users.

These experiments were lead with functional actors of different business processes with the aim to co-construct and validate the language (meta-model and notation) and the involved activities during the modeling process of the ISEA method. They were accomplished with the help of an engineer in Human and Social Sciences Methodologies in charge of the experimental design in MARVELIG, a hotbed scientific experimental platform of Grenoble Informatics Laboratory, for prototype deployment and test in pervasive computing. These experiments are based on user-centered design concepts and attitudes measurement research like [24,35,41]. They were carried out to co-construct and evaluate the ISEA method from a qualitative point of view with two main objectives:

- to co-construct, with the functional actors of the process, the DSL and modeling process that will be the most suited to them,
- to have functional actors' opinion on the support tool ISEAsy.

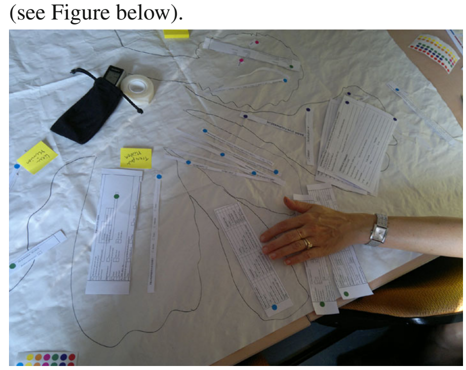
As an illustration, we present in more details two examples of experiments that we lead with two different goals:

- co-construction of the language of the informational perspective. In the cycle presented in Fig. 18, this corresponds to the co-construction of the concrete syntax of the informational perspective during analysis.
- validation of the tool ISEAsy for the organizational perspective. In the cycle presented in Fig. 18, this corresponds to the validation of the tool ISEAsy (for the organizational perspective) during operationalization.

5.1 Experimentation for the co-construction of the language of the informational perspective during analysis

When the goal is to co-construct, with end-users, the DSL or the modeling process the most suited to them, the experiments are generally lead on the form of short-time brainstorming sessions, where we propose a set of activities to be conducted by the participants themselves. During such experiments, the goal for us is to get a consensus on the way to conduct the modeling process or on the concepts to use during the modeling process.

As an example, we present here an experiment that we lead with the goal to get the most suitable notation for the informational perspective. We proposed a set of activities and asked the participants for their opinion in the notation they would like to use to represent concepts of the business process.

Goal	To obtain a consensus on the best representation of the informational perspective
Context	One session of 2 hours 4-6 functional actors of the business process
Pre-conditions	Participants already participated to the organizational perspective elicitation, or have a good knowledge of the business process
Inputs	The organizational perspective is available. The documents are printed.
Proposed activities	The following activities are proposed to the participants: "Cut the documents": participants are invited to cut, in each document, fragments of documents that represent informations shared by the same concept. "Let's play!": participants are invited to play with different domain notations that we propose to them in order to find the best consensual way to group the different fragments in order to represent concepts of the business process. These propositions for domain notations were carried by ourselves during the exploration phase (Fig. 18). In this experiment, we proposed for example to use the idea of houses (a house being a concept and a level being an element of this concept), of towns composed of different houses (a house being a concept and roads between houses being relationships between concepts), …
Results	The expected result is to get the best consensual notation for the informational perspective. In this experiment, the different representations that we proposed were not approved by the end-users who finally proposed the idea of the tree of concepts (see Figure below). 
Consolidation	Once the consensus obtained (during the co-construction phase) on the best notation to use to represent the interactional perspective, the next step for us is to consolidate this idea. As shown in Fig. 18, the design phase should then be raised with new experiments in order to validate, with other end-users, the notion of the tree of concepts. Then, the operationalization phase will have to implement and validate the corresponding tool supporting the tree of concept (see Fig. 16)

We are currently in the operationalization phase for the informational perspective. Indeed, the informational perspective language (dictionary, notation and abstract syntax) and the elicitation activities dedicated to this perspective are validated. But the tool supporting the language and

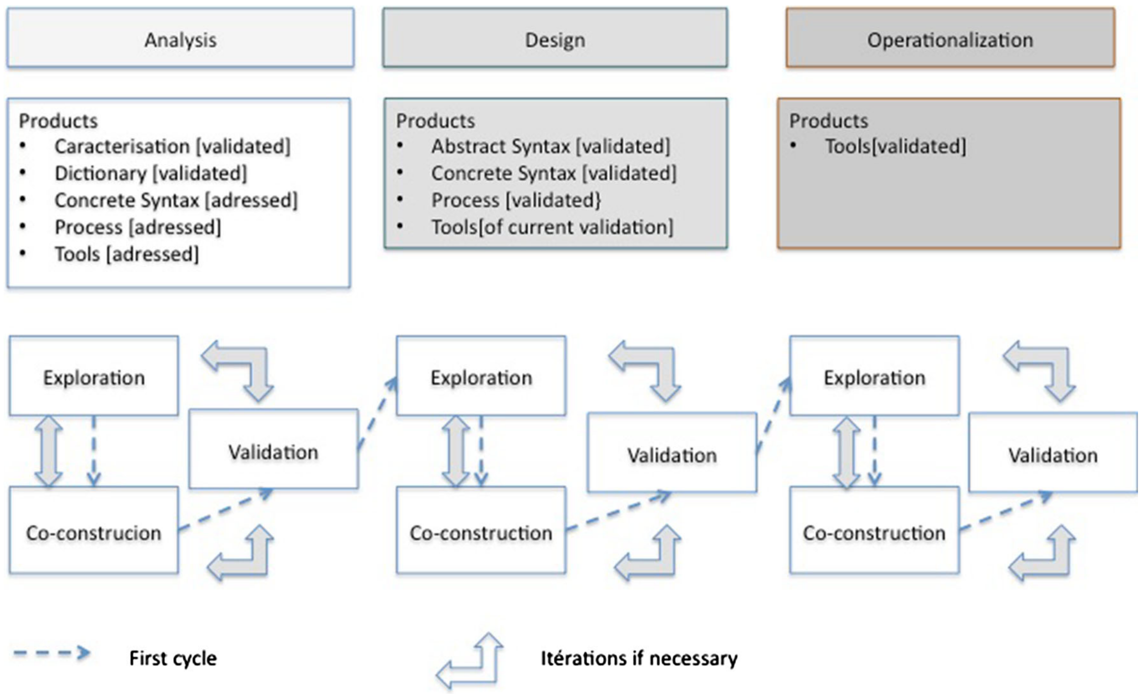


Fig. 18 User-centered validation cycle for co-construction and validation of a method

the evaluation/transformation activities are on current validation. The results of the experiments made to validate this part of the tool ISEAsy will be presented in further works.

5.2 Experimentation for the validation of the tool ISEAsy (organizational perspective) during operationalization

The organizational perspective language (dictionary, notation and abstract syntax) and the modeling process dedicated to this perspective are validated with end-users. In particular, the tool ISEAsy was validated with around twenty business processes. We present here an example of such experiment which goal was to validate the tool (in particular its organizational perspective) from a qualitative point of view, and to measure the usability of the tool.

5.2.1 Experimental protocol

The experiment was lead on a two hours session, aimed to model a particular process in terms of activities and document exchanges, using the ISEA method with the use of ISEAsy. In this experiment, the chosen process was “Welcome of a new employee.” The following table shows the different steps of the protocol.

Steps	Description	Duration
Training on ISEA method	Brief presentation on the ISEA method	5 min
Training on ISEAsy tool	Brief presentation on the tool ISEAsy by an illustration of its main functionalities	10 min
Testing the tool	Realization by the participants of a short practical exercise, in which they are asked to create an action, to create a document, to pass their turn, etc.	15 min
Playing with the tool for the chosen process	Start the elicitation of the organizational perspective for the process “Welcome of a new employee”	80 min
Filling in a satisfaction questionnaire	Fill in a satisfaction questionnaire in order to give opinion on the usability of the tool	10 min
Total time		120 min

5.2.2 Satisfaction questionnaire

A satisfaction questionnaire was distributed to the participants at the end of the experiment. This questionnaire was composed of 27 questions (19 closed questions and 8 open questions). Its purpose is to gather the subjects’ opinion regarding the use and usability of the ISEA method and its support tool ISEAsy. Sample of asked questions are given below:

- The use of post-its to represent activities was:
  very easy—rather easy—rather not easy—not easy—no opinion
- The use of timer events was:
  very easy—rather easy—rather not easy—not easy—no opinion
- Do you think this tool can help you to save time?
  yes—rather yes—rather no—no—no opinion
- Would you be ready to use this tool in a professional context?
- yes—rather yes—rather no—no—no opinion
- Do you think that this tool needs a training phase?
  yes—rather yes—rather no—no—no opinion
- Did this exercise allow you to better understand the business process? Why?
- In your opinion, which are the advantages and inconvenients of this tool? Why?
- …

5.2.3 Participants' profile

The subjects in this experiment were six functional actors involved in the process of welcoming a new employee in the LIG lab: an employee, a team assistant, a team leader, the lab director, the manager of information systems and a human resource assistant. The subjects were not familiar with the method and tool. They didn't participate in previous experiments.

The analysis of the subjects' habits in terms of process description revealed that three of six participants were not used to describe processes. In addition, the three participants which were used to describe processes were not using tools for process description; participant 1 said for example: "I write and draw the main steps of the process in a notebook."

5.2.4 Experiment results

- Participants satisfaction

The results showed that users were satisfied of the practice and the process description obtained. They declared: "the exercise allows to point the dysfunctions and the roles of each of us", "the business process could be clearly described, well detailed and the roles of each were well defined", "each of us has his own role and intervene in the process when needed by taking into account remarks and exchanges". They also added that "the tool was very easy to apprehend" and "allows to formalize the steps of the business process and to have a good visualization of the role of each". The subjects thought that the method and the tool could help them to save time (5 of 6 subjects) and to avoid errors (6 subjects). Moreover, six subjects would be ready to describe the tool's use to a third party.

- Usability of the representation elements

Regarding the usability of the representation elements, participants said that the easiest was to "drag and drop the post-it". But on the contrary, the least easy was to "go get the arrow" and "delete a post-it (click "yes" before validating)".

Overall, the results demonstrated the ease of use of the representation elements: Six of six subjects found it easy to use post-its, five of six subjects (1 no opinion) found easy the document creation, four of six subjects considered easy the tracking of documents (1 subject rather considered not easy and one no opinion) and finally, five of six subjects considered easy the link between activities.

- Participative modeling

The participants considered that the representation of a process in a playful way was rewarding and a good idea in order to remember the work done. They thought that collaborative work can "facilitate communication" and that "with all stakeholders, it is more realistic and more aware of situations that may be different"

5.3 Conclusion

The multiple user-centered experiments lead us to different evolutions of the language and the modeling process. In particular during analysis and design stages, the language was considerably simplified in order to be usable by the end-users. We suppressed a lot of elements that we firstly thought useful for the end-users to model the business processes: conditions, repetitive actions, actions composing an activity, etc. At the end, the DSL proposed in the organizational perspective contains very few simple elements: activity, external actor, document, temporal event, and recursive action, end of participation and change of actor (see Figs. 4, 5). We consider now that the organizational perspective language (dictionary, notation and abstract syntax) and the modeling process dedicated to this perspective are validated.

Concerning the informational perspective, the language (dictionary, notation and abstract syntax) and the elicitation activities dedicated to this perspective are validated. The tool supporting the language and the evaluation/transformation activities are currently under validation.

Finally, the interactional perspective is yet in co-construction. If the dictionary and the concrete syntax are validated, the abstract syntax is under definition. The tool supporting the language and the elicitation activities are only addressed: They are a prototype built with existing tools such as Balsamiq and Gambit, allowing us to lead experiments. Figure 19 shows examples of hand-made sketches using the domain model to help the participants to design the desired sketches.

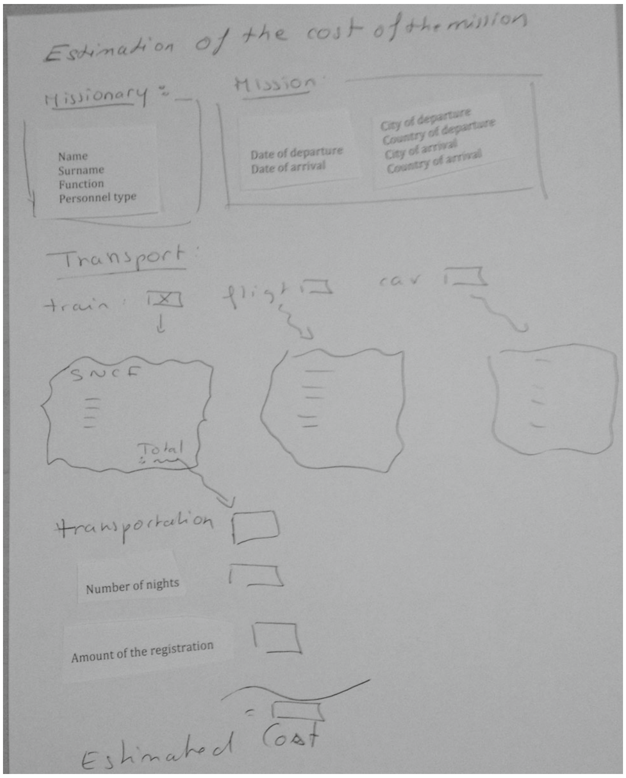


Fig. 19 Co-construction of the interactional perspective: a "hand-made" screen

6 Conclusion and further works

ISEA is an end-user participative method that proposes a multi-perspective business process modeling and improvement. The modeling process is defined as a map where the goals are to elicit and improve end-users business process models before their automation and to construct analysis models in standard languages. The strategies between the goals are realized by participative and playful activities.

The ISEA method was inspired from participative methods, but also from traditional information systems methods. In particular, the decomposition of activities in atomic actions in the organizational perspective was inspired from the MERISE method [37] the cutting activity and the resulting document fragments in the elicitation of the informational perspective can be compared to the Message Structures technique of Communication Analysis [16] and the navigation between screens in the interactional perspective are directly inspired from the UsiXML approach [40]. Moreover, as already said, the modeling languages are inspired and pruned from standard modeling languages (BPMN for the organizational perspective and UML for the informational perspective).

In this study, ISEA allows modeling with three perspectives: organizational, informational and interactional. All the proposed strategies have not the same maturity degree. The strategies for the elicitation/evaluation/transformation of the organizational perspective were the results of several evaluations. ISEAsy, the support tool of ISEA method, was used since 2011 for the elicitation and improvement of the UPMF Grenoble University business processes. Moreover, in the context of the fusion of Grenoble universities, the ISEA method is used since September 2014 to unify the business processes of the three different universities. The tool integrates a basic transformation into BPMN, the resulting models are accessible with the BPM tool Bonita.

Proposals for the informational perspective are in the operationalization stage: The tool integrated in ISEAsy was the object of demonstrations, which allowed improving the elicitation of the informational perspective. However, the validation experiments remain in attempt. The proposals for the interactional perspective were only evaluated with a restricted set of users (only one process with 6 participants).

As further results, the first purpose is to complete the evaluation of models and strategies of the informational and interactional perspectives, and to complete the map with other perspectives, for example, intentional or decisional perspectives that are particularly useful for elicitation of innovative processes. The second purpose is to take into account new types of processes: right now, ISEA is suited to existing administrative processes: we have started a study to evaluate the usability of ISEA on co-design processes in industrial organizations. We are convinced that the two purposes "new business processes perspectives" and "new types of processes" are linked. For example, intentional and decisional perspectives will be essential to elicit innovative processes. During the previous experiments on the university processes, we first proposed an intentional perspective aimed to identify process goals. Nevertheless, in the context of our experiments (existing processes which need to be improved), this step didn't seem to be useful to the participants who wanted to focus on their daily activities and on the encountered problems.

Long-term perspectives are to apply ISEA to other domains than business process management, more particular for knowledge acquisition in the context of innovative collaborative projects. As the next step of this research, we will indeed focus on innovation processes and knowledge acquisition during innovation projects. Our ANR ACIC Project (The French National Research Agency-Absorptive Capacity in Companies) [1] started in January 2015, will aim to integrate the process improvement with absorption capacity notion, and find out how this notion can affect the improvement of innovation process. The result of these future studies will be presented as an extension of ISEA method.

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